

# A numerical investigation of SNR shortfall of preamplifier decoupling

Wenjun Wang, Rasmus Alexander Jepsen, Lars G. Hanson,  
Vitaliy Zhurbenko, Jan Henrik Ardenkjær-Larsen

September 24, 2024

## Synopsis

Preamplifier noise complicates in-array coil matching. However, the popular matching scheme, preamplifier decoupling, does not yield the highest SNR. In this abstract, the SNR shortfall of preamplifier decoupling is studied numerically using electromagnetic simulation models. We show that the highest SNR can be unreachable in practice. When the highest SNR is reachable, preamplifier decoupling yields SNR close to the maximum as long as low-noise preamplifiers are used.

## 1 Introduction

In recent years, preamplifier decoupling [1] has been questioned as inessential [2], [3] or detrimental to achieving optimal SNR [4], [5] because it does not account for preamplifier noise from other channels. Array noise matching (ANM) [4], [5] is proposed to correct for cross-channel preamplifier noise and reach the highest possible SNR.

In this abstract, we investigate when ANM is feasible for coil arrays, and the shortfall of preamplifier-decoupled image SNR relative to ANM for different array channel counts, sizes, and preamplifiers.

## 2 Theory

Our model is shown in Figure 1(a). Denote  $\mathcal{P}$  as a preamplifier with its matching network. Let  $\mathcal{P}$  have  $N_L$  as its Lange's  $N$  [6] and  $F_{\text{opt}}$  as its minimum noise factor.  $\mathcal{P}$ 's optimal noise impedance  $Z_{\text{opt},\mathcal{P}} = R_{\text{opt},\mathcal{P}} + jX_{\text{opt},\mathcal{P}}$  is adjusted by matching.

A phantom is placed inside an  $N$ -channel coil array, shown in Figure 2(a–d). The  $l$ -th coil is connected to preamplifier  $\mathcal{P}_l$ ,  $1 \leq l \leq N$ . A voxel induces signal voltage on each channel, written as a vector  $\mathbf{u} = [u_1, u_2, \dots, u_N]^T$ . For each voxel, after optimal weighting [1], [7], an optimal power SNR is reached, named “O-1 SNR” [4]:

$$\text{SNR}_{\text{O1}} := \mathbf{u}^H (\mathbf{A} + \mathbf{B})^{-1} \mathbf{u}, \quad (1)$$

where

$$\mathbf{A} = 4k_B T \cdot \text{BW} \cdot F_{\text{opt}} \left( \mathbf{Z}_a + \mathbf{Z}_a^H \right), \quad (2)$$

$$\mathbf{B} = (\mathbf{Z}_a - [\mathbf{Z}]) \left[ \frac{2P_{\text{opt};\infty}}{R_{\text{opt},\mathcal{P}_l}} \right] (\mathbf{Z}_a - [\mathbf{Z}])^H, \quad (3)$$

$$P_{\text{opt};\infty} = 4k_B T \cdot \text{BW} \cdot N_L, \quad (4)$$

where  $[\cdot]^H$  denotes Hermitian conjugation;  $[\dots]$  are diagonal matrices;  $[Z]$  is diagonal with the  $l$ -th element  $([Z])_{l,l} = Z_{\text{opt},\mathcal{P}_l}$ . For preamplifier decoupling, we let  $([Z])_{l,l} = (\mathbf{Z}_a)_{l,l}$ .  $\mathbf{A}$ ,  $\mathbf{B}$  can be interpreted as the minimal and added array noise figure, respectively.

In some situations ANM is feasible. After ANM and optimal weighting, preamplifier noise is corrected for the particular voxel. The SNR reaches another optimum there, named “O-2 SNR” [4]:

$$\text{SNR}_{\text{O2}} := \mathbf{u}^H \mathbf{A}^{-1} \mathbf{u}, \quad (5)$$

reached when

$$([Z])_{l,l} = \left( \frac{\mathbf{u}^H \mathbf{A}^{-1} \mathbf{Z}_a}{\mathbf{u}^H \mathbf{A}^{-1}} \right)_l. \quad (6)$$

We call a voxel “O-2 admissible” if the optimal noise resistances of all  $\mathcal{P}_l$  are positive, i.e.,  $\Re([Z])_{l,l} > 0$ , for all  $1 \leq l \leq N$ .

ANM can be done only for one voxel. After ANM, we get an SNR map where the target voxel reaches O-2 SNR, and others reach O-1. When ANM is performed for the image center, the resulting SNR map is called an “O-2c SNR map”.

If all preamplifiers are noiseless, a further optimal SNR can be reached, named “O-3 SNR”. The relation between “O-1 SNR”, “O-2 SNR”, “O-2c SNR” and “O-3 SNR” is shown in Figure 1(b).

### 3 Methods

The setup is shown in Figure 2(a). A phantom is surrounded by an 8-, 12-, or 16-channel array of  $\varnothing 268$  mm,  $\varnothing 342.56$  mm, or  $\varnothing 417.12$  mm as shown in Figure 2(b–d). The arrays are terminated by one of four preamplifiers: noiseless, Elcry2-u [8], Infineon BFP740 and BFR35AP [9]. Their noise parameters are listed in Figure 2(f). The frequency is 63.89 MHz.

The method is sketched in Figure 1(c). All arrays are optimally overlapped at first. The impedance matrix and H-field of each in-array coil element are simulated. The H-field on a slice in Figure 2(e) is exported. A self-developed software computes voxels’ signal vectors as described by Hoult [10], generates the O-1 SNR maps of preamplifier-decoupled arrays, and determines whether a voxel is O-2 admissible. If the image center is admissible, the O-2c SNR maps of preamplifier terminations in Figure 2(f), and the difference between O-1 SNR and O-2c SNR, are computed.

### 4 Results

As shown in Figure 3, only a small region near the slice center is O-2 admissible, which shrinks as an array gets a larger size or more channels, and finally vanishes for the 12-channel and 16-channel  $\varnothing 417.12$  mm arrays. The maximal, minimal, average O-1 and O-2c SNRs are drawn in Figure 4. For small-sized arrays, the O-1 SNRs of Elcry2-u and BFP740 are not far from those of noiseless amplifiers. From Figure 4(c), for 8- and 12-channel arrays below  $\varnothing 342.56$  mm, the SNR improvement by ANM is below 40%, or 1.4 dB. SNR improvements over 50% are achieved when a high-noise-figure preamplifier is used, or an array has a larger size or more channels. The former case is avoided in real life, and the latter involves admissibility below 0.4%.

The O-1 and O-2c SNR maps of the 16-channel,  $\varnothing 268$  mm array is shown in Figure 5. Other arrays show similar patterns. For low-noise preamplifiers the ANM improves SNR only marginally. Although ANM improves the SNR of the BFR35AP-terminated array, the resulting SNR is not as high as Elcry2-u’s and BFP740’s, as also shown in Figure 4(b) and Figure 5(b).

## 5 Conclusion

Preamplifier-decoupled O-1 SNR is not far from ANM's O-2c SNR when ANM is practically feasible and low-noise preamplifiers are used.

In most practical cases, preamplifier decoupling only takes a small toll on image SNR, but greatly simplifies coil tuning. Preamplifier decoupling should remain the choice for coil builders.

## Figures and Tables

Figure 1: (a) A model of coil array terminated by preamplifiers.  $u_1, u_2, \dots, u_N$  are the signal a voxel generates on each coil. (b) The relation between O-1, O-2 and O-3 SNRs. For a particular array geometry, at a particular voxel, O-1 SNR can vary depending on the preamplifier terminations, but O-2 SNR and O-3 SNR are fixed numbers. ANM can only be performed for one voxel because it is done at the hardware level. (c) A flow chart of our method.  $\mathbf{M} = [1, j, 0]/\sqrt{2}$  and  $\mathbf{M} = [1, -j, 0]/\sqrt{2}$  give the same results throughout the article.

Figure 2: (a) The simulation setup consisting of a phantom, an array, and a shield. Array: 230 mm long, diameter (dashed) 268 mm, 342.56 mm, or 417.12 mm. Shield:  $\varnothing 680$  mm, length 840 mm; phantom:  $\varnothing 200$  mm, length 230 mm,  $\epsilon_r = 76.82$ ,  $\sigma = 1.233$  S/m. Frequency 63.89 MHz. (b) 8-, (c) 12-, and (d) 16-channel arrays. (e) The slice for SNR investigation, in the middle of the phantom, 115 mm from both ends. (f) Preamplifiers' noise parameters, all in linear scale.

Figure 3: O-2 admissible regions of each array. The region of interest is within the black circle. Gray: O-2 inadmissible; White: O-2 admissible. The 12- and 16-channel arrays of  $\varnothing 417.12$  mm have the whole region of interest inadmissible.

Figure 4: SNR statistics. All SNRs are in terms of power. (a) Relative O-1 SNRs of preamplifier decoupling. Each relative SNR equals the ratio of SNR to the central image SNR of the same array channel count and size with noiseless amplifiers so that noiseless preamplifiers give unit SNR at image center. (b) The resulting relative SNRs of ANM for the image center. For the 12- and 16-channel arrays of  $\varnothing 417.12$  mm, the entire image is inadmissible. (c) The SNR improvements by ANM for the image center. The 16-channel arrays have their maximal and central SNRs overlapped.

Figure 5: SNR maps of a 16-channel,  $\varnothing 268$  mm array. The reference point at which SNR = 1 is marked by +. All SNRs are in terms of power.  $\sigma$ : standard deviation. The SNR improvements of Elcry2-u and BFP740 are below 40%.

## References

- [1] P. B. Roemer, W. A. Edelstein, C. E. Hayes, S. P. Souza, and O. M. Mueller, "The NMR phased array," *Magnetic Resonance in Medicine*, vol. 16, no. 2, pp. 192–225, Nov. 1990. [Online]. Available: <http://doi.wiley.com/10.1002/mrm.1910160203>.

- [2] A. Reykowski, C. Saylor, and G. R. Duensing, “Do We Need Preamplifier Decoupling?” In *Proceedings of the 19th Annual Meeting of the International Society of Magnetic Resonance in Medicine*, Montréal, QC, Canada, 2011, p. 3883.
- [3] C. Findelee, O. Lips, P. Vernickel, C. Leussler, and R. Duensing, “Preamp decoupling improves SNR and the earth is flat,” in *Proceedings of the 27th Annual Meeting of the International Society of Magnetic Resonance in Medicine*, Montréal, QC, Canada, 2019, p. 563.
- [4] C. Findelee, “Array noise matching- Generalization, proof and analogy to power matching,” *IEEE Transactions on Antennas and Propagation*, vol. 59, no. 2, pp. 452–459, Feb. 2011.
- [5] C. Findelee, “Array Noise Matching via the Scattering Matrix,” *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 4, pp. 2344–2353, 2019.
- [6] J. Lange, “Noise Characterization of Linear Twoports in Terms of Invariant Parameters,” Tech. Rep. 2, 1967.
- [7] S. Applebaum, “Adaptive arrays,” *IEEE Transactions on Antennas and Propagation*, vol. 24, no. 5, pp. 585–598, Sep. 1976. [Online]. Available: <http://ieeexplore.ieee.org/document/1141417/>.
- [8] *ElCry Electronics*. [Online]. Available: <http://elcry.com/>.
- [9] Keysight Technologies, *E4440A PSA Spectrum Analyzer, 3 Hz to 26.5 GHz*. [Online]. Available: <https://www.keysight.com/us/en/product/E4440A/psa-spectrum-analyzer-3-hz-to-265-ghz.html>.
- [10] D. Hoult and P. C. Lauterbur, “The sensitivity of the zeugmatographic experiment involving human samples,” *Journal of Magnetic Resonance (1969)*, vol. 34, no. 2, pp. 425–433, May 1979. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/0022236479900192>.